

DAYAPURAM RESIDENTIAL SCHOOL

SHORT NOTE – QUADRILATERALS

- A quadrilateral is a 4-sided polygon that has 4 angles too.
- The sum of the interior angles of a quadrilateral is 360°

RECTANGLE:

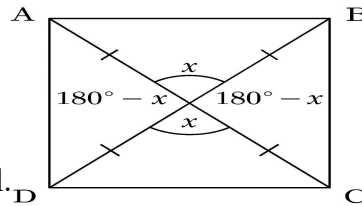
A rectangle is a quadrilateral with 4 angles 90° and opposite sides are equal.

Properties:

1. The opposite sides are parallel.
2. The diagonals are equal and intersect at their midpoint.

Angles and sides properties.

- The vertically opposite angles are equal.
- The diagonals are equal.
- The angle opposite to equal sides are equal.
- Linear pair: the pair of angles together forms a line, and the sum is 180° .

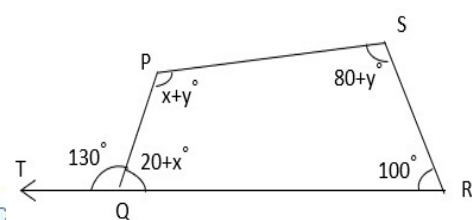
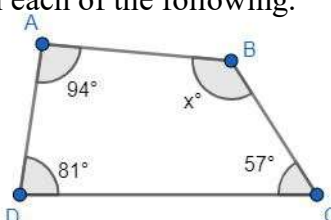
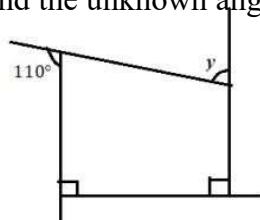


Construction of a Rectangle:

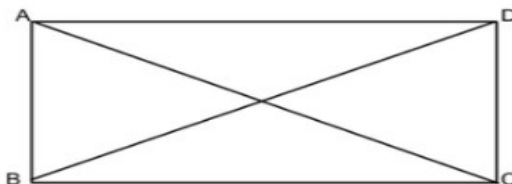
- If the diagonal and the angle between the diagonals are given, we can construct the rectangle by drawing the bisector with the given angle.

QUESTIONS:

1. The opposite angles of a rectangle are: _____
2. The sum of the measures of the interior angles of any quadrilateral is
a) 180° b) 360° c) 270° d) 540°
3. One angle of a quadrilateral is 150° and the other three angles are equal. What is the measure of each of these equal angles?
(a) 75° (b) 85° (c) 95° (d) 70°
4. If two angles are supplementary and one angle is double the other, then the larger angle is:
a) 60 b) 120 c) 45 d) 30
5. Find the unknown angle in each of the following.



6. Consider the rectangle $ABCD$, $AC = 2x + 1$ and $BD = 3x - 4$. Find the value of x .



7. Prove that the diagonals of a rectangle are equal.
8. Prove that the diagonals of a rectangle bisect each other at their midpoint.
9. Construct a rectangle with diagonals 5cm and an angle between them as 60° using the diagonal properties of rectangle.

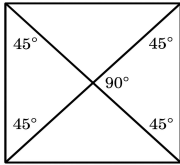
SQUARE:

- A square is a quadrilateral with all sides equal and all angles 90° .
- Every square is a rectangle.

Properties of a Square:

- The opposite sides are parallel.
- The diagonals of a square bisect each other at 90° .
- The diagonal bisects the angles of the square.
- The diagonals are equal.

The angles of a square:



Construction of a square:

- If the diagonal is given, draw the bisector to construct the square.

Questions:

1. If the diagonals of a quadrilateral bisect each other at 90° , it will be a ____
a) Rectangle b) Square c) Parallelogram d) Kite
2. Prove that the diagonals of a square bisect each other at 90° .
3. Prove that the diagonals of a square are equal.
4. Draw a square with a diagonal of 6cm using the properties of square.

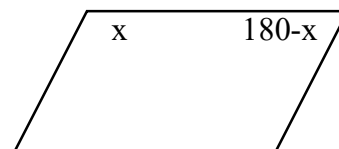
Parallelogram:

- A parallelogram is a quadrilateral with parallel opposite sides.
- Every rectangle is a parallelogram.

Properties of a parallelogram:

- The opposite sides of a parallelogram are equal.
- The diagonals of a parallelogram bisect each other.

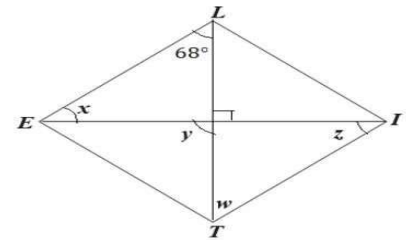
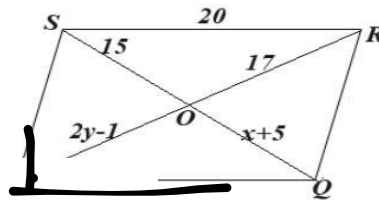
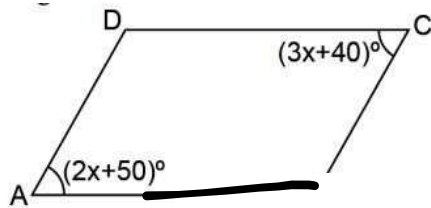
The angles of a parallelogram:



- Adjacent angles of a parallelogram add up to 180° .
- The opposite angles are equal.
- **Construction of a parallelogram:**
- If the adjacent sides and the angle between them are given, then we can draw the parallelogram.

Questions:

1. In a parallelogram, the adjacent angles are:
 - a) Equal b) Supplementary c) Complementary d) None of these
2. In a parallelogram, the opposite angles are:
 - b) Equal b) Supplementary c) Complementary d) None of these
3. Find the value of x, y, z in the following parallelogram.



4. Construct a parallelogram with sides 6 cm and 5 cm and angle between them as 60°
5. Prove that the opposite sides of a parallelogram are equal.

Rhombus:

- A quadrilateral in which all the sides have the same length is a rhombus.
- Every Rhombus is a parallelogram.
- Every square is a rhombus.

Properties of a Rhombus:

- Opposite sides are parallel.
- The diagonals bisect each other at 90°
- The diagonals bisect the angle of the Rhombus.

Angles in the Rhombus:

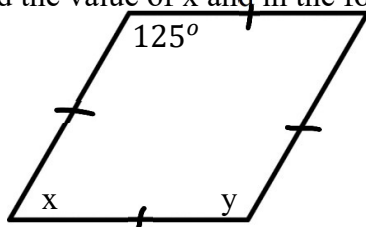
- The opposite angles are equal.
- Adjacent angles are supplementary.

Construction of a Rhombus:

- If the diagonals are given, draw the bisector to construct a Rhombus.

Questions:

- In a Rhombus, the opposite angles are:
a) Equal b) Supplementary c) Complementary d) None of these
- Construct a Rhombus with diagonals 6cm and 5 cm using the diagonal properties of the Rhombus.
- If the diagonals of a quadrilateral is perpendicular to each other, it will be a ____
a) Rectangle b) Rhombus c) Parallelogram d) Kite
- Find the value of x and y in the following.



- Prove that the diagonals of a rhombus are perpendicular to each other.

Kite:

- A kite is a quadrilateral with two pairs of adjacent sides equal.

Trapezium:

- A trapezium is a quadrilateral with one pair of parallel opposite sides.
➤ Every parallelogram, rhombus, rectangle, and square is a trapezium.

Angles of a trapezium:

- The sum of adjacent angles of non-parallel sides is supplementary.

Isosceles Trapezium:

- When the non-parallel sides are equal for a trapezium, then it is called an Isosceles Trapezium.

Angles:

- The angles opposite to equal sides are equal.

Question:

- Find the all the other angles:

